Naive Bayes Classifier – Part 1: Conditional Probability

Before jumping into Naive Bayes, we need to understand **Conditional Probability**, since it's the foundation of the algorithm.

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1 What is Probability?

- Probability measures **how likely an event is**.
  Example:
- Probability of getting heads when tossing a coin = $P(\text{Heads}) = 0.5.$

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2 What is Conditional Probability?

Conditional probability = probability of an event happening **given that another event has already happened**.

Mathematically:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Where:

- $P(A|B)$ = Probability of **A happening given B happened**.
- $P(A \cap B)$ = Probability that **both A and B happen**.
- $P(B)$ = Probability that **B happens**.

👉 Example:

- Let event A = student is a girl.
- Event B = student passed exam.
- $P(A|B)$ = Probability that a student is a girl **given** that they passed.

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3 Why is this important for Naive Bayes?

Naive Bayes is based on **Bayes' Theorem**, which uses conditional probability:

$$P(Class|Data) = \frac{P(Data|Class) \cdot P(Class)}{P(Data)}$$

Here:

- $P(Class|Data)$ → Probability of a class given the data (posterior).
- $P(Data|Class)$ → Likelihood of data given the class.
- $P(Class)$ → Prior probability of the class.
- $P(Data)$ → Normalizing constant.

So, **conditional probability** is the tool that allows us to flip the perspective: instead of asking

👉 "What's the chance of data given class?"

We ask

👉 "What's the chance of class given data?"

That's exactly how classifiers like Naive Bayes work.

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4 Intuition with Example

Imagine you're classifying emails: Spam or Not Spam.

- P(Spam|"Buy Now") = Probability that the email is spam **given** it contains the word "Buy Now."
- To compute this, we use conditional probability.

➤ Quick Recap

- **Conditional Probability** tells us the probability of an event given another event.
  Formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$.
- It's the **foundation** for Bayes' theorem, which powers Naive Bayes Classifier.
